

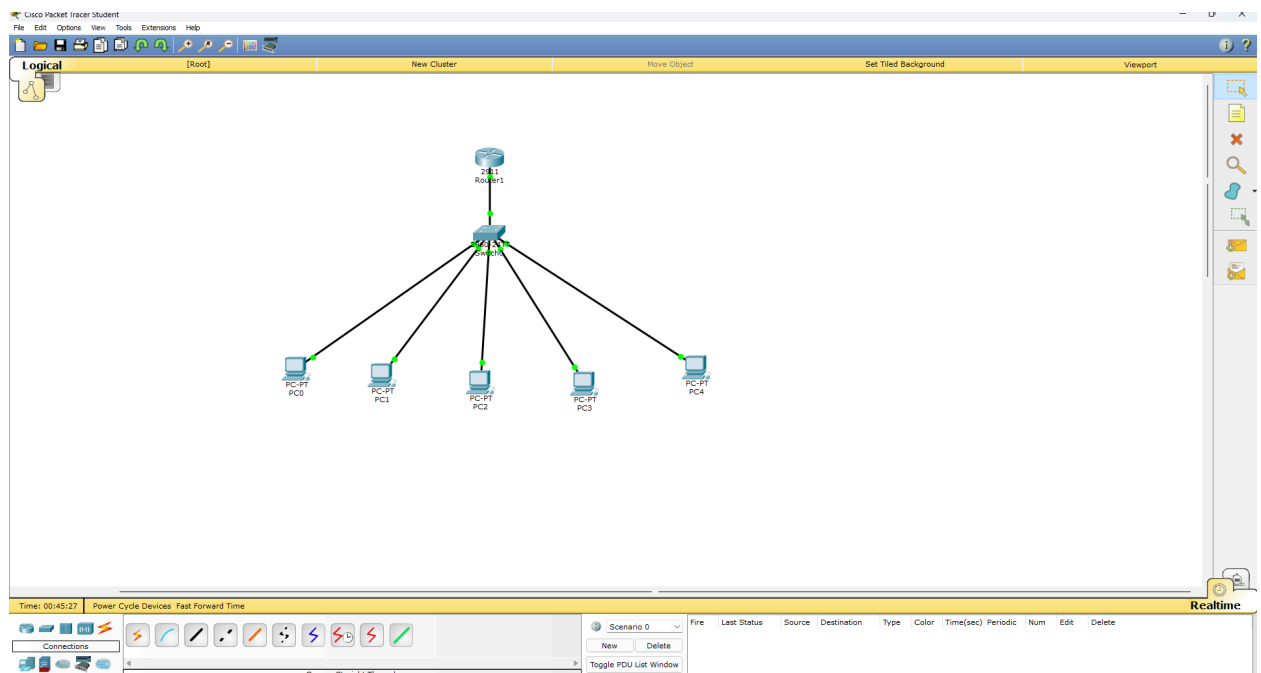
Experiment 9 – Implementation of Subnetting in Cisco Packet Tracer

Aim:

To implement subnetting using Cisco Packet Tracer.

Screenshots:

Subnetting Network Topology in Cisco Packet Tracer:



Network Configuration:

PC0

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IP Address: 192.168.1.11

Subnet Mask: 255.255.255.224

Default Gateway: 192.168.1.1

DNS Server:

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address: /

Link Local Address: FE80::202:17FF:FE8D:8DD6

IPv6 Gateway:

IPv6 DNS Server:

PC1

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IP Address: 192.168.1.12

Subnet Mask: 255.255.255.224

Default Gateway: 192.168.1.1

DNS Server:

IPv6 Configuration

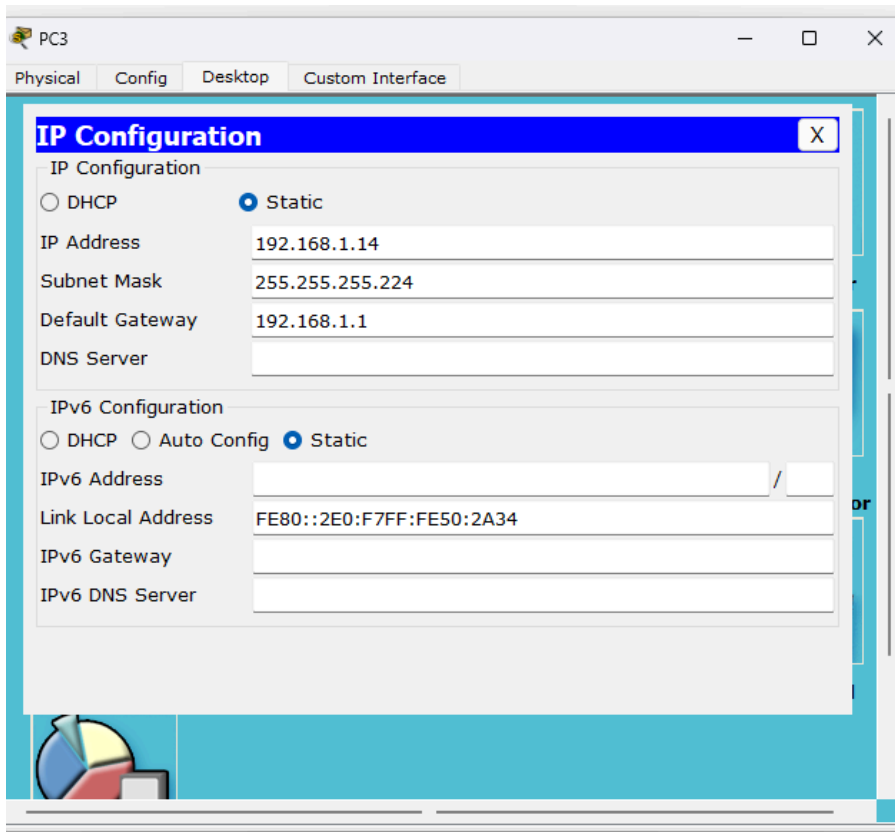
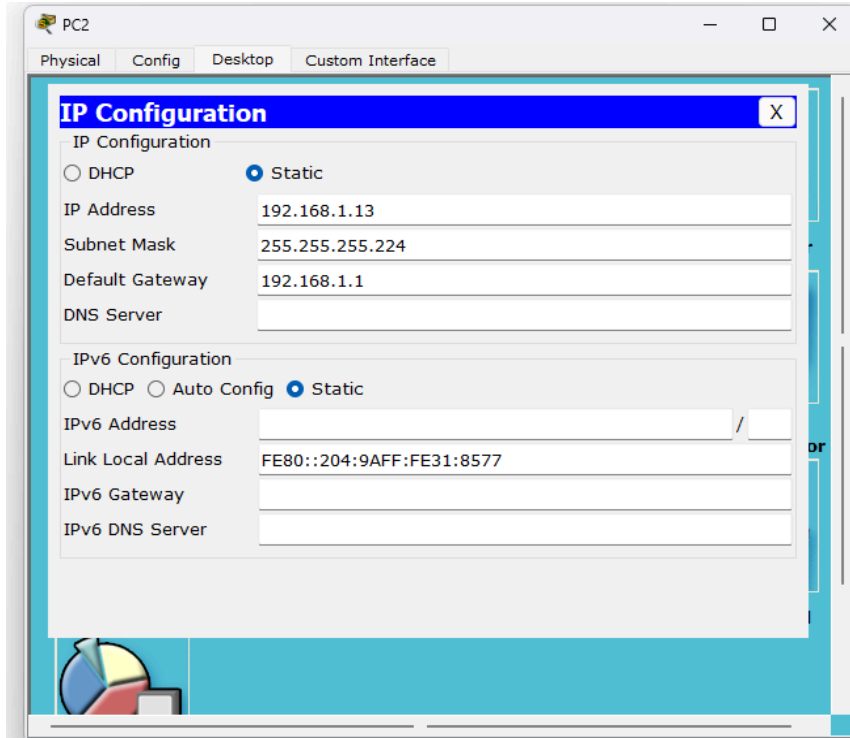
☐ DHCP ☐ Auto Config ☒ Static

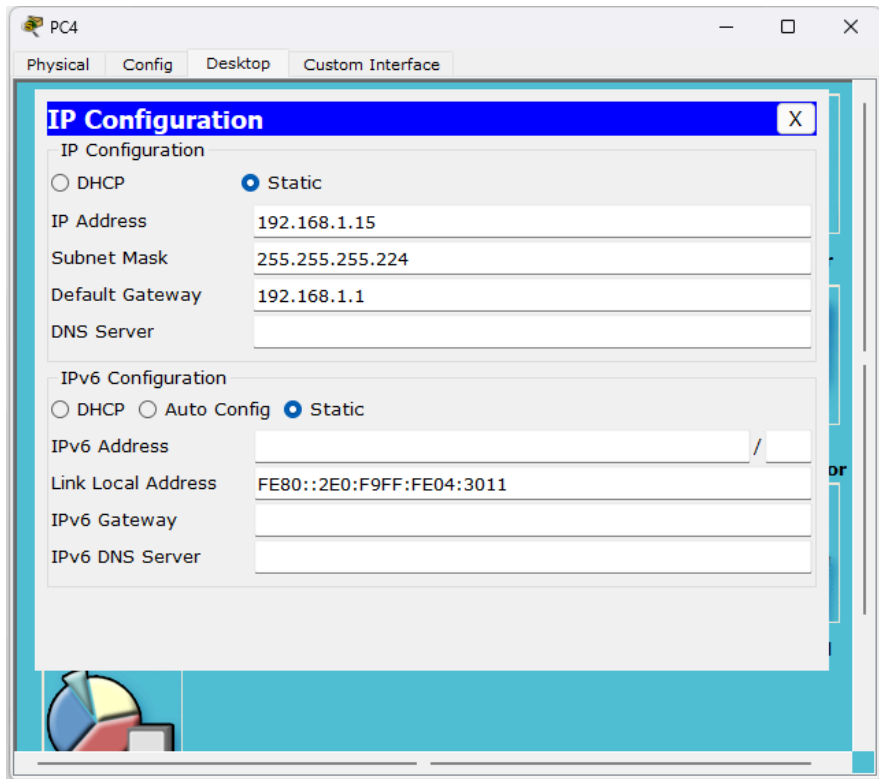
IPv6 Address: /

Link Local Address: FE80::2E0:8FFF:FE2A:6A53

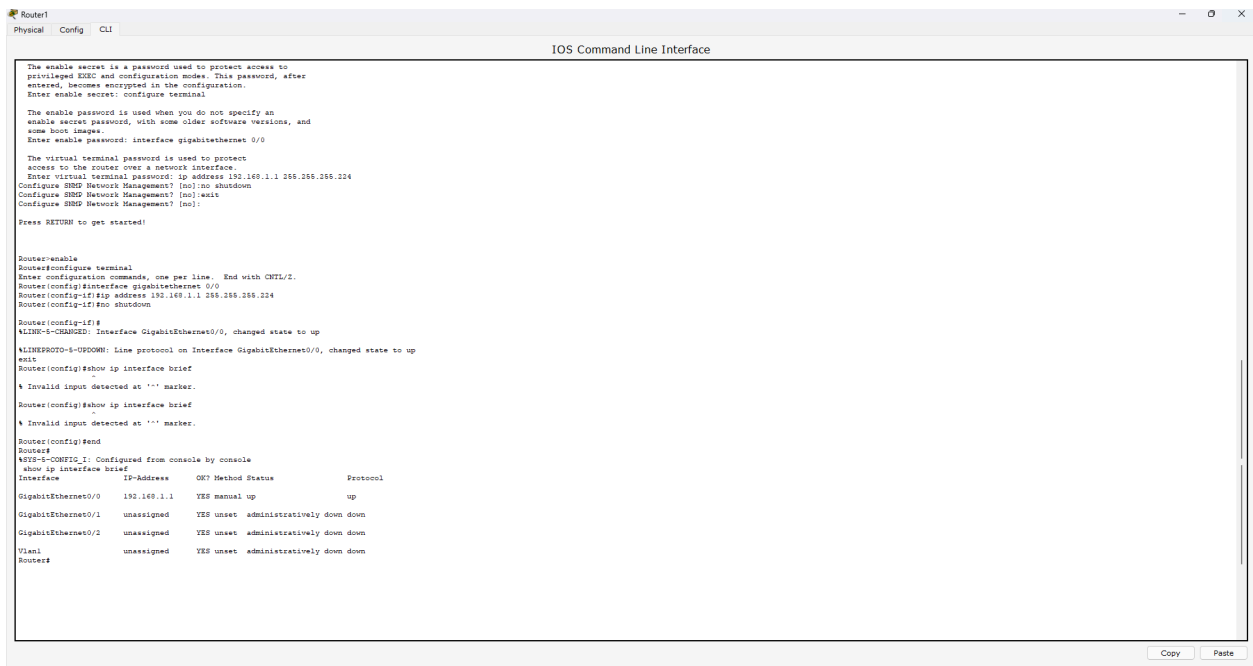
IPv6 Gateway:

IPv6 DNS Server:





Router Configuration:



Ping test:

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Packet Tracer PC Command Line 1.0
PC>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=0ms TTL=255
Reply from 192.168.1.1: bytes=32 time=0ms TTL=255
Reply from 192.168.1.1: bytes=32 time=0ms TTL=255
Reply from 192.168.1.1: bytes=32 time=0ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
PC>ping 192.168.1.12

Pinging 192.168.1.12 with 32 bytes of data:

Reply from 192.168.1.12: bytes=32 time=0ms TTL=128
Reply from 192.168.1.12: bytes=32 time=0ms TTL=128
Reply from 192.168.1.12: bytes=32 time=0ms TTL=128
Reply from 192.168.1.12: bytes=32 time=0ms TTL=128

Ping statistics for 192.168.1.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
PC>ping 192.168.1.13

Pinging 192.168.1.13 with 32 bytes of data:

Reply from 192.168.1.13: bytes=32 time=0ms TTL=128
Reply from 192.168.1.13: bytes=32 time=0ms TTL=128
Reply from 192.168.1.13: bytes=32 time=0ms TTL=128
Reply from 192.168.1.13: bytes=32 time=0ms TTL=128

Ping statistics for 192.168.1.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
PC>ping 192.168.1.14

Pinging 192.168.1.14 with 32 bytes of data:

Reply from 192.168.1.14: bytes=32 time=0ms TTL=128
Reply from 192.168.1.14: bytes=32 time=0ms TTL=128
Reply from 192.168.1.14: bytes=32 time=0ms TTL=128
Reply from 192.168.1.14: bytes=32 time=0ms TTL=128

Ping statistics for 192.168.1.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
PC>ping 192.168.1.15

Pinging 192.168.1.15 with 32 bytes of data:
```